

MALLIKARJUN

AIML Undergraduate — Machine Learning & AI Developer

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Professional Summary

AIML undergraduate specializing in Machine Learning, Data Analytics, and AI system development. Experienced in the full ML lifecycle, from data preprocessing and feature engineering to model tuning and evaluation. Proven track record of building and deploying predictive models, NLP pipelines, and time-series forecasting systems using REST APIs and LLM integrations.

Education

B.E. in Artificial Intelligence & Machine Learning

Poojya Doddappa Appa College of Engineering

2023–2027

CGPA: 8.18

Pre-University Course (PUC) – Science

Shree Guru Independent PU College

2021–2023

85.16%

Projects

Myra – Real-Time Conversational System (Ongoing)

FastAPI, LLMs, STT/TTS

- Developing a locally deployed real-time system using a modular asynchronous (ASGI) architecture for low-latency interactions.
- Integrating speech-to-text, text-to-speech, and LLM-driven responses to facilitate dynamic, human-like engagement.
- Implemented emotion-aware response logic and interactive virtual character behavior, moving beyond standard chatbot workflows.

Time Series Forecasting (Demand Prediction)

XGBoost, Pandas, NumPy

- Built a forecasting model using lag and rolling feature engineering to predict demand trends.
- Applied time-based train-test split and evaluated performance using MAE, RMSE, and MAPE (5%).
- Conducted trend plots, Actual vs Predicted graphs, and feature importance analysis.

Distributed SIEM Pipeline (ELK + Kafka)

Docker, Kafka, Elasticsearch, Logstash, Kibana, Python

- Architected and containerized an enterprise-grade SIEM pipeline using the ELK stack and Apache Kafka to ingest, parse, and analyze unstructured Syslog data in real-time.
- Implemented a decoupled, high-throughput message broker architecture with Kafka to safely buffer log traffic and prevent data loss during high-volume attack spikes.
- Authored custom Logstash Grok filters to extract structured network fields (IP, username) from raw SSH logs, tag malicious vectors, and map geo-locations in Kibana.
- Developed an automated threat-detection Python script utilizing Elasticsearch APIs to trigger real-time alerts upon detecting ≥ 5 failed login attempts within a 60-second window, reducing MTTR.

AeroCore – Telemetry Monitoring System

Django, ML, ReportLab

- Engineered a full-stack platform to analyze engine telemetry and predict Remaining Useful Life (RUL).
- Architected secure multi-tier authentication, admin workflows, and automated SMTP email notifications.
- Developed dynamic PDF reporting and a scalable backend for real-time data processing and state management.

AI YUGA Club Management Platform

React, Node.js, PostgreSQL

- Engineered a platform for event management, team registrations, scoring, and real-time leaderboards.
- Implemented JWT-based role-access control and automated certificate generation with QR-based verification.
- Managed scalable booking/reporting workflows deployed on Vercel and Render.

Technical Skills

Programming: Python, SQL, JavaScript

Core Concepts: Data Structures, OOPS, Basic Algorithms, Probability, Statistics

Machine Learning: Scikit-learn, TensorFlow/Keras, Classification, Regression, Time Series Forecasting, Model Evaluation, Hyperparameter Tuning

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Microsoft Excel, Data Cleaning, EDA, Feature Engineering

NLP & AI: TF-IDF, Text Vectorization, Large Language Models (LLMs), Prompt Engineering

Backend, Systems & DevOps: Django, FastAPI, Node.js, Express.js, REST APIs, JWT Authentication, LLM Integration, Docker, Apache Kafka

Frontend: HTML, CSS, JavaScript, React.js, Tailwind CSS

Databases & Search Engines: SQLite, PostgreSQL, Elasticsearch

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Streamlit, Logstash, Kibana, Vercel, Render